Lecture 14: Two-Port Circuits

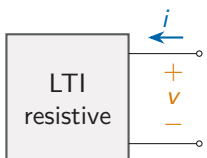
EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

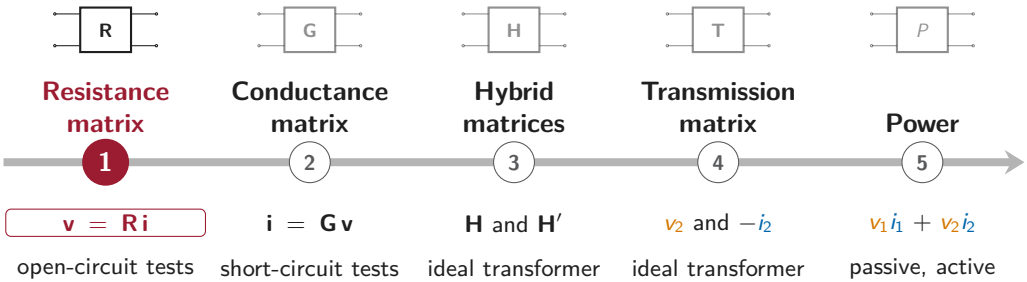
current voltage, node voltage e

Review: one-ports

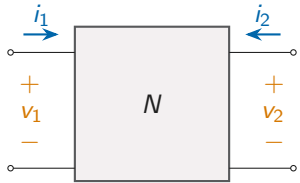


- No independent sources: $v = R_{\text{in}}\, i$
- With independent sources: $v = R_{\text{th}}\, i + v_{\text{th}}$

Roadmap

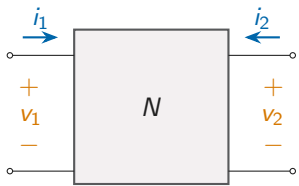

R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
1
2
3
4
5
v = Ri
i = Gv
H and H′
v2 and −i2
v1i1 + v2i2
open-circuit tests
short-circuit tests
ideal transformer
ideal transformer
passive, active

Motivation: two ports



- N: LTI resistive, no independent sources
- An input and an output: an amplifier, a transformer, a filter
- One-port: $v = R\,i$ with scalars
- Two-port: $\mathbf{v} = \mathbf{R}\,\mathbf{i}$ with vectors in $\mathbb{R}^2$ and a 2×2 matrix

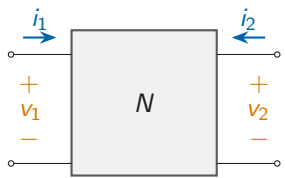
The port variables



- Each port: the current enters at + and leaves at −
- Four variables: v_1, i_1, v_2, i_2
- Two equations: two variables in terms of the other two

given	find	matrix
i_1, i_2	v_1, v_2	resistance **R**
v_1, v_2	i_1, i_2	conductance **G**
i_1, v_2	v_1, i_2	hybrid **H**
v_2, $-i_2$	v_1, i_1	transmission **T**

The resistance matrix



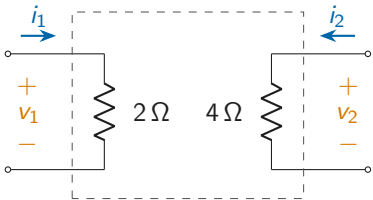
$$v_1 = r_{11}\, i_1 + r_{12}\, i_2$$

$$v_2 = r_{21}\, i_1 + r_{22}\, i_2$$

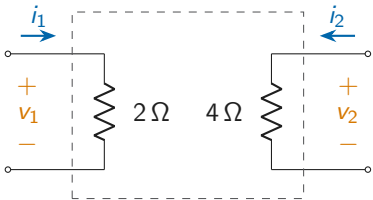
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Example 14.1: two separate resistors

$\mathbf{R} = ?$



Example 14.1: two separate resistors



$\mathbf{R} = ?$

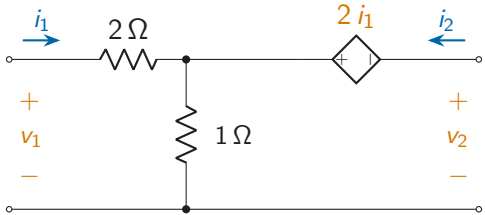
$v_1 = 2\, i_1 + 0 \cdot i_2$

$v_2 = 0 \cdot i_1 + 4\, i_2$

$$\mathbf{R} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \Omega$$

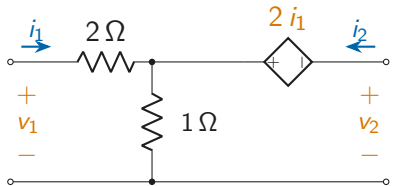
No coupling: $r_{12} = r_{21} = 0$

Example 14.2: a dependent source inside



$$\mathbf{R} = ?$$

Example 14.2: KVL

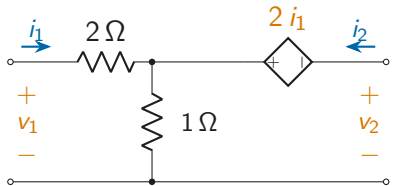


$1\,\Omega$ carries $i_1 + i_2$

$v_1 = 2\,i_1 + (i_1 + i_2) = 3\,i_1 + i_2$

$v_2 = -2\,i_1 + (i_1 + i_2) = -i_1 + i_2$

Example 14.2: KVL



$1\,\Omega$ carries $i_1 + i_2$

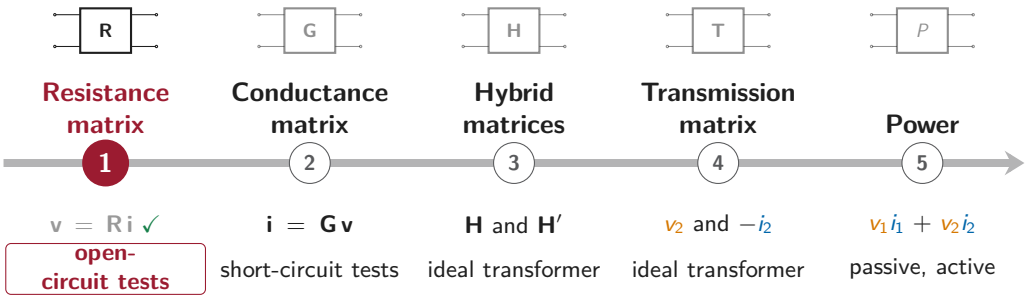
$v_1 = 2\,i_1 + (i_1 + i_2) = 3\,i_1 + i_2$

$v_2 = -2\,i_1 + (i_1 + i_2) = -i_1 + i_2$

$$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \Omega$$

$r_{12} \neq r_{21}$: the dependent source

Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
1
2
3
4
5
v = Ri ✓
open-circuit tests
i = Gv
short-circuit tests
H and H′
ideal transformer
v_2 and $-i_2$
ideal transformer
$v_1 i_1 + v_2 i_2$
passive, active

The entries from two tests

Port 2 open: $i_2 = 0$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_{11}\, i_1 \\ r_{21}\, i_1 \end{bmatrix} \quad \Rightarrow \quad r_{11} = \frac{v_1}{i_1}, \quad r_{21} = \frac{v_2}{i_1}$$

the first column: r_{11} input resistance, r_{21} transfer resistance

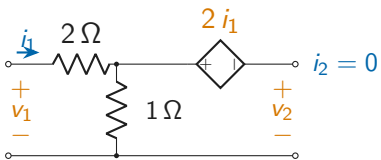
Port 1 open: $i_1 = 0$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} 0 \\ i_2 \end{bmatrix} = \begin{bmatrix} r_{12}\, i_2 \\ r_{22}\, i_2 \end{bmatrix} \quad \Rightarrow \quad r_{12} = \frac{v_1}{i_2}, \quad r_{22} = \frac{v_2}{i_2}$$

the second column: r_{12} transfer resistance, r_{22} input resistance

Example 14.2 revisited: the two open-circuit tests

Port 2 open

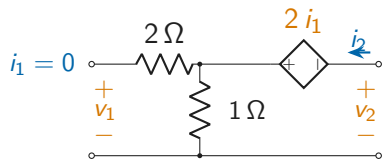


No current through the dependent source

$v_1 = 3\,i_1 \;\Rightarrow\; r_{11} = 3\,\Omega$

$v_2 = -2\,i_1 + i_1 \;\Rightarrow\; r_{21} = -1\,\Omega$

Port 1 open

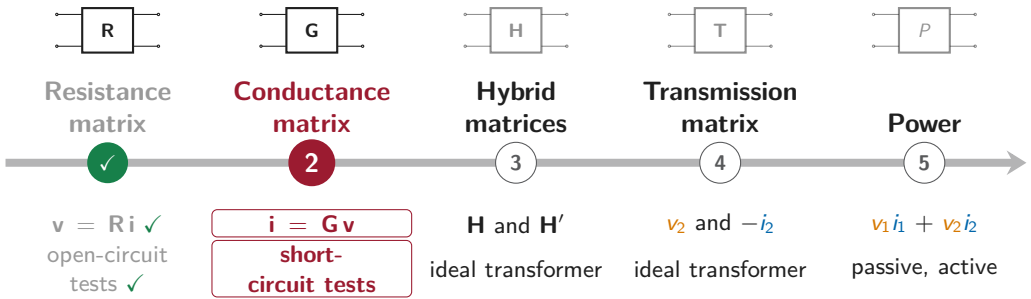


No current in 2 Ω; the dependent source is 0

$v_1 = i_2 \;\Rightarrow\; r_{12} = 1\,\Omega$

$v_2 = i_2 \;\Rightarrow\; r_{22} = 1\,\Omega$ The same **R** ✓

Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
✓
2
3
4
5
v = Ri ✓
open-circuit tests ✓
i = Gv
short-circuit tests
H and H′
ideal transformer
v_2 and $-i_2$
ideal transformer
$v_1 i_1 + v_2 i_2$
passive, active

The conductance matrix

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}}_{\mathbf{G}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

When $\mathbf{R}^{-1}$ exists: $\mathbf{G} = \mathbf{R}^{-1}$

The entries from two short-circuit tests

Port 2 shorted: $v_2 = 0$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}\, v_1 \\ g_{21}\, v_1 \end{bmatrix} \quad \Rightarrow \quad g_{11} = \frac{i_1}{v_1}, \quad g_{21} = \frac{i_2}{v_1}$$

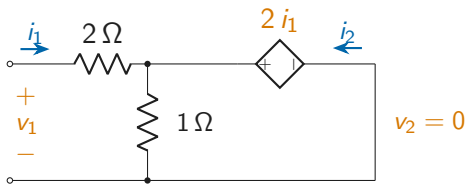
the first column: g_{11} input conductance, g_{21} transfer conductance

Port 1 shorted: $v_1 = 0$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} g_{12}\, v_2 \\ g_{22}\, v_2 \end{bmatrix} \quad \Rightarrow \quad g_{12} = \frac{i_1}{v_2}, \quad g_{22} = \frac{i_2}{v_2}$$

the second column: g_{12} transfer conductance, g_{22} input conductance

Example 14.3: G for Example 14.2



Port 2 shorted: $v_2 = 0$

Mesh 1: $-v_1 + 2\,i_1 + (i_1 + i_2) = 0$

Mesh 2: $-2\,i_1 + (i_1 + i_2) = 0$

Mesh 2: $i_2 = i_1$

Mesh 1: $v_1 = 4\,i_1$

$g_{11} = \frac{1}{4}\,\mathrm{S}, \quad g_{21} = \frac{1}{4}\,\mathrm{S}$

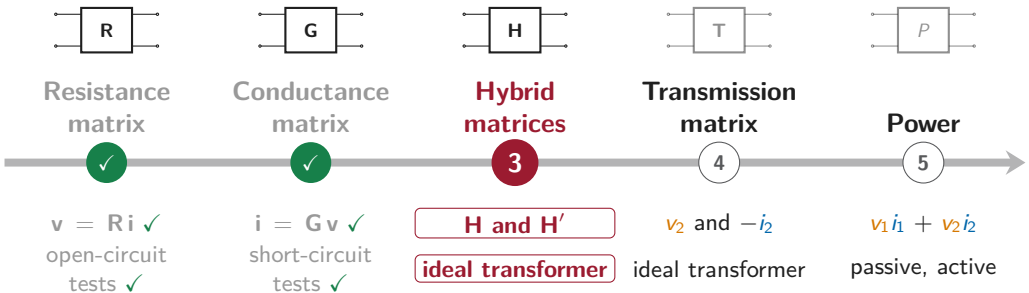
Example 14.3: the result

The second test ($v_1 = 0$) gives the second column

$$\mathbf{G} = \mathbf{R}^{-1} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \mathrm{S}$$

R or **G** need not exist

Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
✓
✓
3
4
5
v = Ri ✓
open-circuit tests ✓
i = Gv ✓
short-circuit tests ✓
H and H′
ideal transformer
v2 and −i2
ideal transformer
v1i1 + v2i2
passive, active

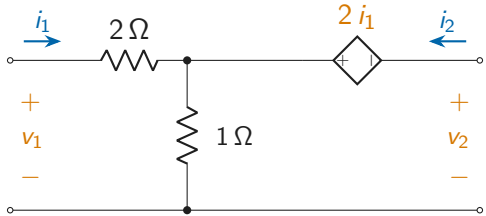
Hybrid matrices

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \mathbf{H}' \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

- $\mathbf{H}' = \mathbf{H}^{-1}$ when the inverse exists
- h_{11}, h_{21}: port 2 shorted, $v_2 = 0$
- h_{12}, h_{22}: port 1 open, $i_1 = 0$

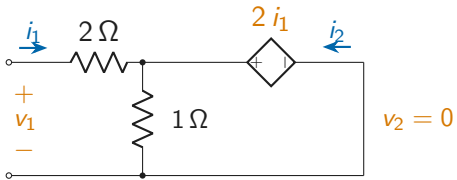
Example 14.4: H of a circuit



The circuit of Example 14.2: $\mathbf{H} = ?$

Example 14.4: two tests

Port 2 shorted: $v_2 = 0$



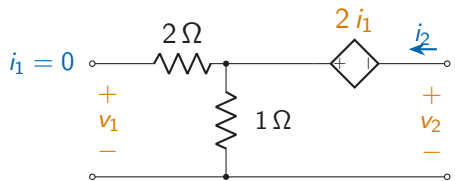
Mesh 2: $-2\,i_1 + (i_1 + i_2) = 0$

$\Rightarrow \quad i_2 = i_1$

Mesh 1: $v_1 = 2\,i_1 + (i_1 + i_2) = 4\,i_1$

$h_{11} = 4\,\Omega, \quad h_{21} = 1$

Port 1 open: $i_1 = 0$



The dependent source is 0: $v_2 = i_2$

$\Rightarrow \quad i_2 = v_2$

$v_1 = i_2 = v_2$

$h_{12} = 1, \quad h_{22} = 1\,\mathrm{S}$

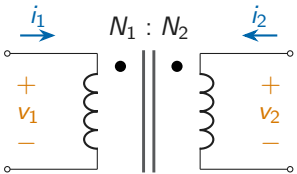
Example 14.4: the result

$$\mathbf{H} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$$

- Units: h_{11} in Ω, h_{22} in S, h_{12} and h_{21} without units
- Check with $\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$ from Example 14.2 ✓

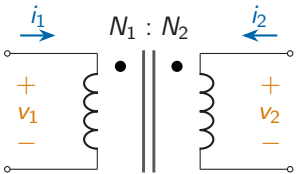
Example 14.5: the ideal transformer

$\mathbf{H} = ?$ $\mathbf{R} = ?$



$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \quad N_1\, i_1 + N_2\, i_2 = 0$$

Example 14.5: the ideal transformer



$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \qquad N_1\, i_1 + N_2\, i_2 = 0$$

$\mathbf{H} = ?$ $\mathbf{R} = ?$

With $n = N_1/N_2$:

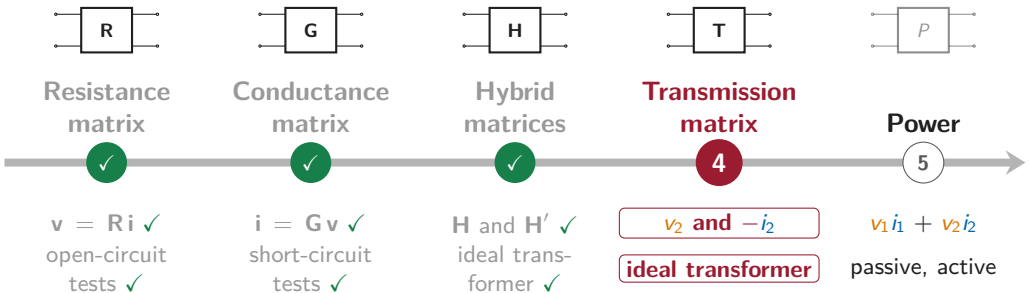
$v_1 = n\, v_2, \qquad i_2 = -n\, i_1$

$$\mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$$

Ports open: $i_1 = i_2 = 0$, but v_1, v_2 are free

No $\mathbf{R}$, no $\mathbf{G}$

Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
4
5
v = Ri ✓
open-circuit tests ✓
i = Gv ✓
short-circuit tests ✓
H and H′ ✓
ideal transformer ✓
v2 and −i2
ideal transformer
v1 i1 + v2 i2
passive, active

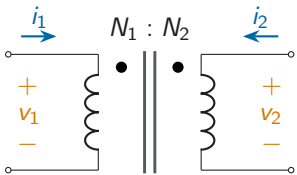
The transmission matrix

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

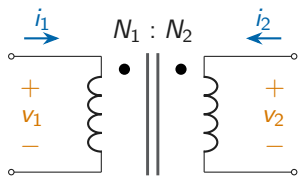
- Also called the chain parameters
- Port 1 in terms of port 2
- $-i_2$: the current that leaves port 2

Example 14.6: T of the ideal transformer

$$\mathbf{T} = ?$$



Example 14.6: T of the ideal transformer



$\mathbf{T} = ?$

$$v_1 = \frac{N_1}{N_2}\, v_2$$

$$i_1 = -\frac{N_2}{N_1}\, i_2 = \frac{N_2}{N_1}\,(-i_2)$$

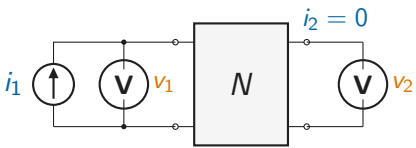
$$\mathbf{T} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix}$$

Which test? Open or short

Matrix	Inputs	1st column	2nd column
R	i_1, i_2	port 2 open	port 1 open
G	v_1, v_2	port 2 **shorted**	port 1 **shorted**
H	i_1, v_2	port 2 **shorted**	port 1 open
H′	v_1, i_2	port 2 open	port 1 **shorted**
T	v_2, $-i_2$	port 2 open	port 2 **shorted**

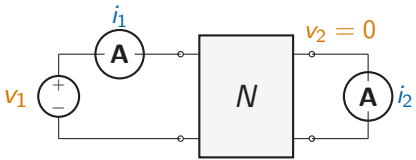
Zero the other input: $i = 0 \Rightarrow$ open, $v = 0 \Rightarrow$ **short**

Measuring a two-port



Open-circuit test

$$r_{11} = \frac{v_1}{i_1}, \quad r_{21} = \frac{v_2}{i_1}$$



Short-circuit test

$$g_{11} = \frac{i_1}{v_1}, \quad g_{21} = \frac{i_2}{v_1}$$

- Ideal voltmeter: an open; ideal ammeter: a **short**
- One meter reading divided by the source value

Example 14.7: R from two measurements

	i_1	i_2	v_1	v_2
Exp. 1	1	1	5	5
Exp. 2	1	-1	3	-1

No port open: $\mathbf{R} = ?$

Example 14.7: R from two measurements

	i_1	i_2	v_1	v_2
Exp. 1	1	1	5	5
Exp. 2	1	-1	3	-1

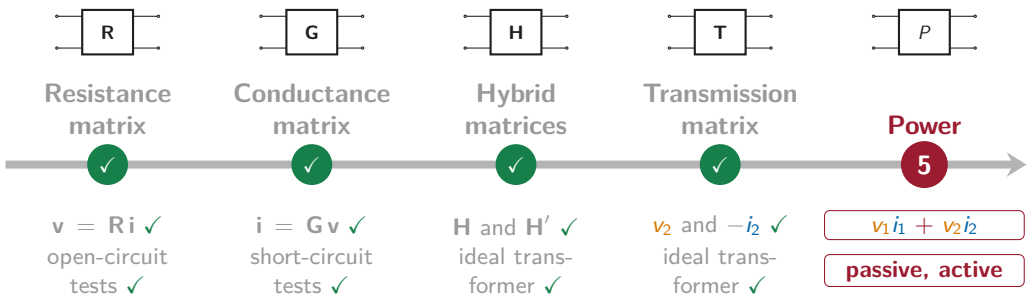
No port open: $\mathbf{R} = ?$

$$\begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} = \mathbf{R} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

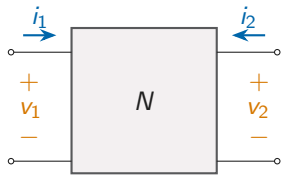
$$\mathbf{R} = \mathbf{V}\,\mathbf{I}^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Any two independent experiments

Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
5
v = Ri ✓
open-circuit tests ✓
i = Gv ✓
short-circuit tests ✓
H and H′ ✓
ideal transformer ✓
v2 and −i2 ✓
ideal transformer ✓
v1i1 + v2i2
passive, active

The power of a two-port



P_N: the sum of the powers of all elements inside N

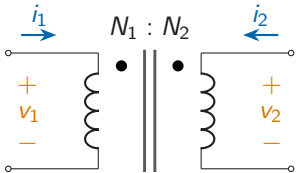
$$P_N = v_1\, i_1 + v_2\, i_2$$

Proof: Tellegen's theorem (previously)

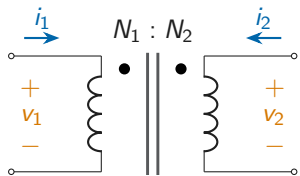
- **Passive:** $P_N \geq 0$ for all possible port variables
- **Active:** not passive

Example 14.8: is the ideal transformer passive?

$$P_N = v_1\, i_1 + v_2\, i_2$$



Example 14.8: is the ideal transformer passive?



$P_N = v_1\, i_1 + v_2\, i_2$

Plug $v_1 = n\, v_2$, $i_2 = -n\, i_1$:

$P_N = n\, v_2\, i_1 - n\, v_2\, i_1 = 0$

$P_N = 0$: passive

It neither absorbs nor delivers power

Summary

Matrix	Equation	Tests
resistance **R**	$[v_1 \ v_2]^T = \mathbf{R}\,[i_1 \ i_2]^T$	ports open
conductance **G**	$[i_1 \ i_2]^T = \mathbf{G}\,[v_1 \ v_2]^T$	ports shorted; $\mathbf{G} = \mathbf{R}^{-1}$
hybrid **H**	$[v_1 \ i_2]^T = \mathbf{H}\,[i_1 \ v_2]^T$	port 2 shorted, port 1 open
transmission **T**	$[v_1 \ i_1]^T = \mathbf{T}\,[v_2 \ -i_2]^T$	port 1 from port 2

$$P_N = v_1\, i_1 + v_2\, i_2; \ \text{ passive if } P_N \geq 0 \text{ always}$$